

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 07 — EXPLORING ICS PACKET CAPTURES

Student Edition

Authorized ICDFA training environments only
icdfa.edu.ng

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Date	01-10-2026

Lab 07: Exploring ICS Packet Captures

Course: Intro to Critical Infrastructure for IT Learners | Estimated Duration: 4–5 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Use Wireshark display filters to analyze provided ICS packet captures.
- Recognize HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7 communication.
- Identify device, protocol and credential-exposure indicators in recorded traffic.
- Explain security weaknesses visible in legacy or cleartext industrial communications.
- Produce a concise packet-analysis report without interacting with any external system.

Laboratory Environment

System	Expected Address	Account	Password
Analyst/Kali VM	172.16.2.5 (teaching image)	kali	kali

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

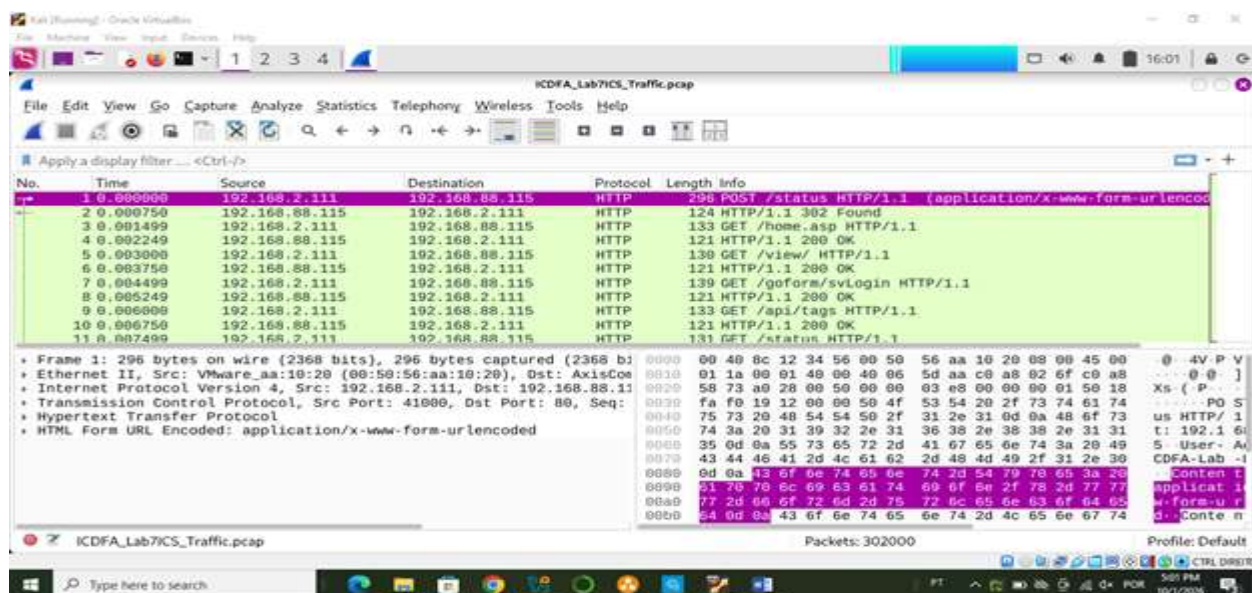
Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

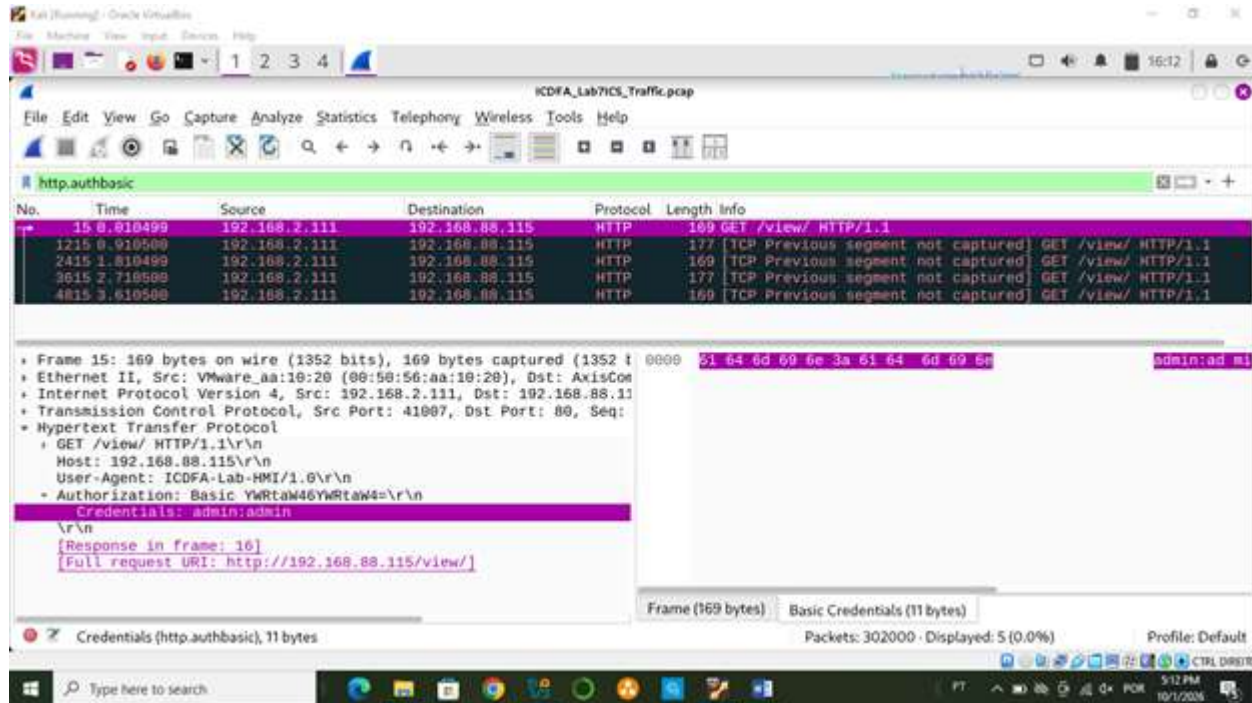
Procedure

Step 1. Start the ICDFA analyst VM in an isolated environment. This lab uses instructor-provided packet captures only; no scanning, exploitation or credential reuse is required.

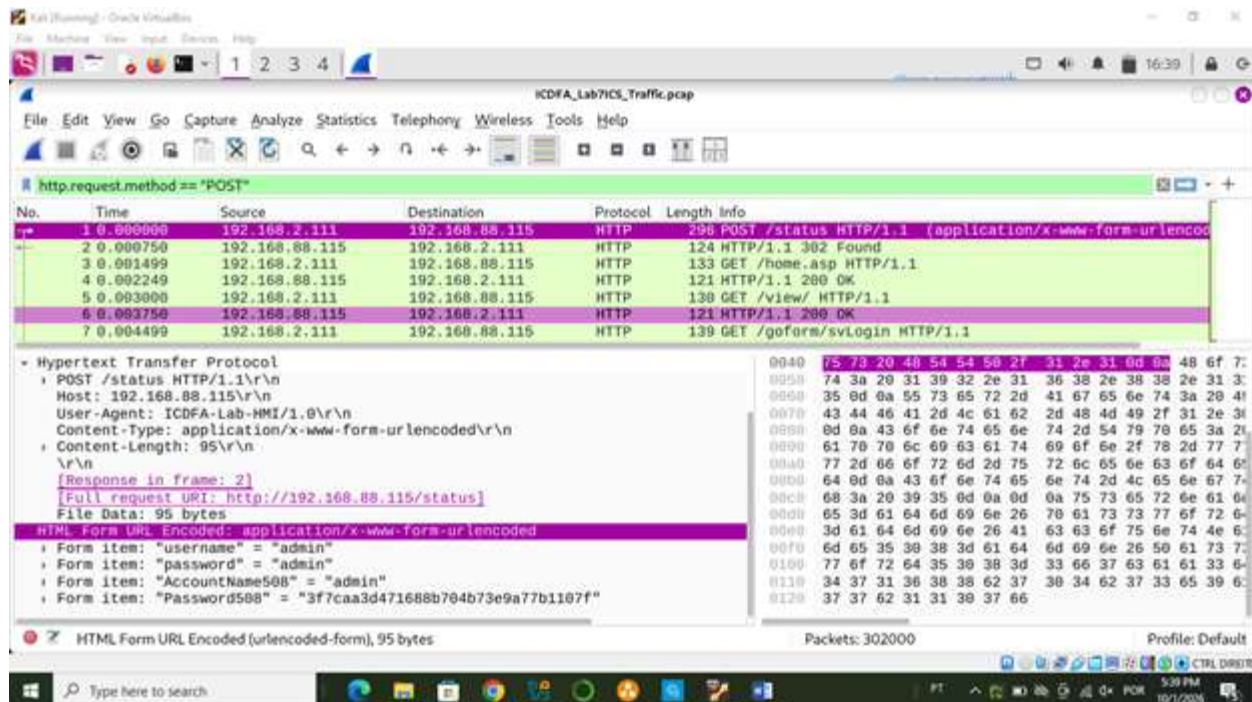
Step 2. Open Wireshark and load the first instructor-provided ICS capture (for the teaching dataset this may be 4SICS-GeekLounge-151021.pcap).



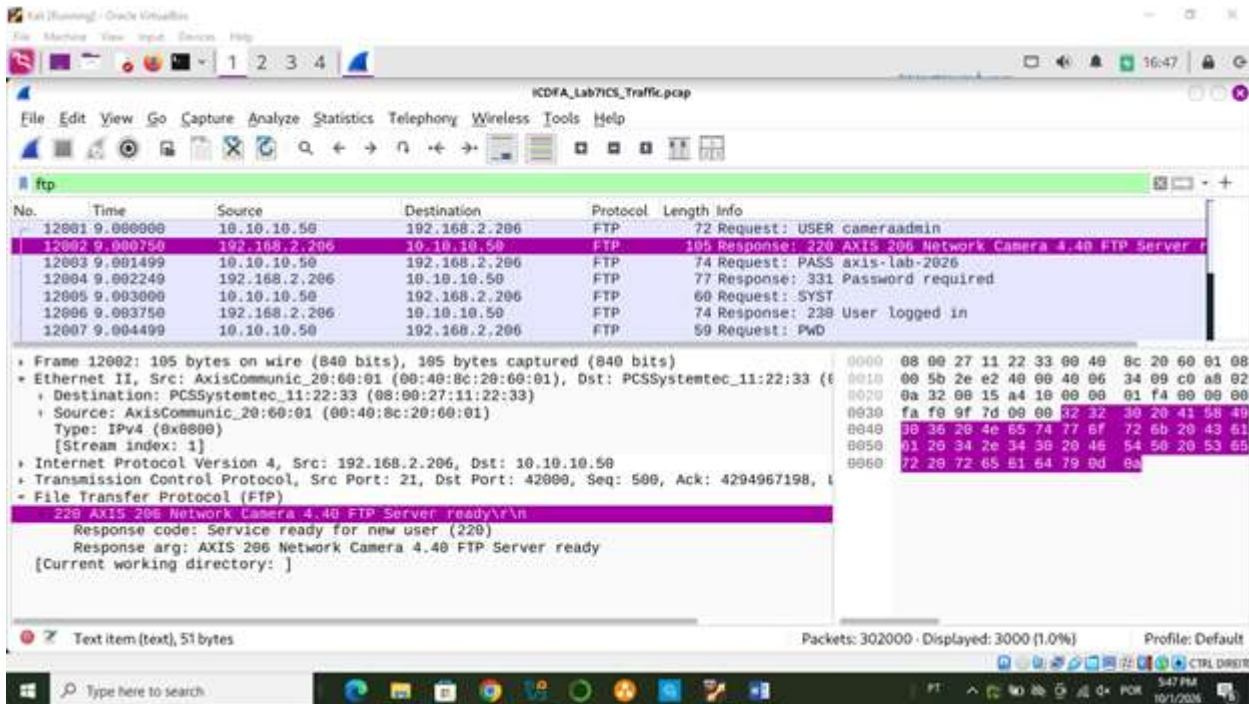
Step 3. Apply the filter `http.authbasic`. Inspect the HTTP Authorization field and determine whether Basic authentication material is present. If you decode an encoded value for learning purposes, do so only within the provided capture and redact the recovered credential from submitted screenshots.



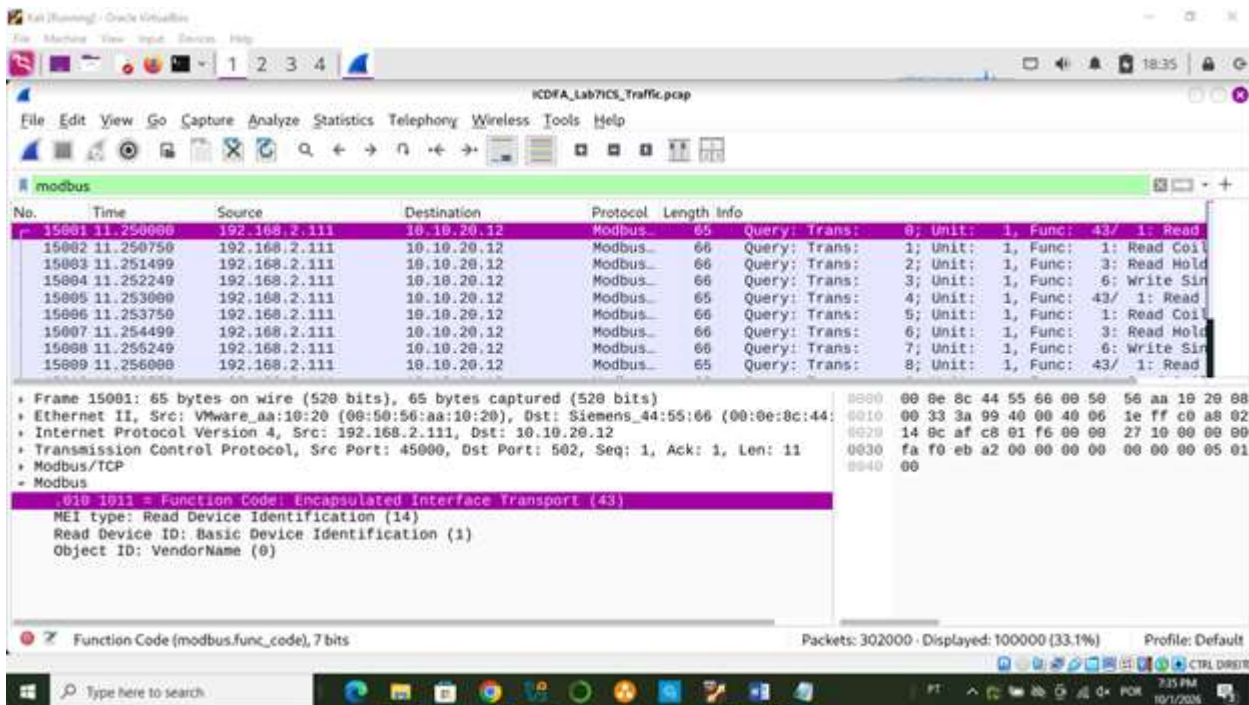
Step 4. Clear the filter and apply `http.request.method == "POST"`. Refine to `http.request.method == "POST"` and inspect form submissions, request paths and response status codes. Record security-relevant observations without attempting to log in to any identified device.



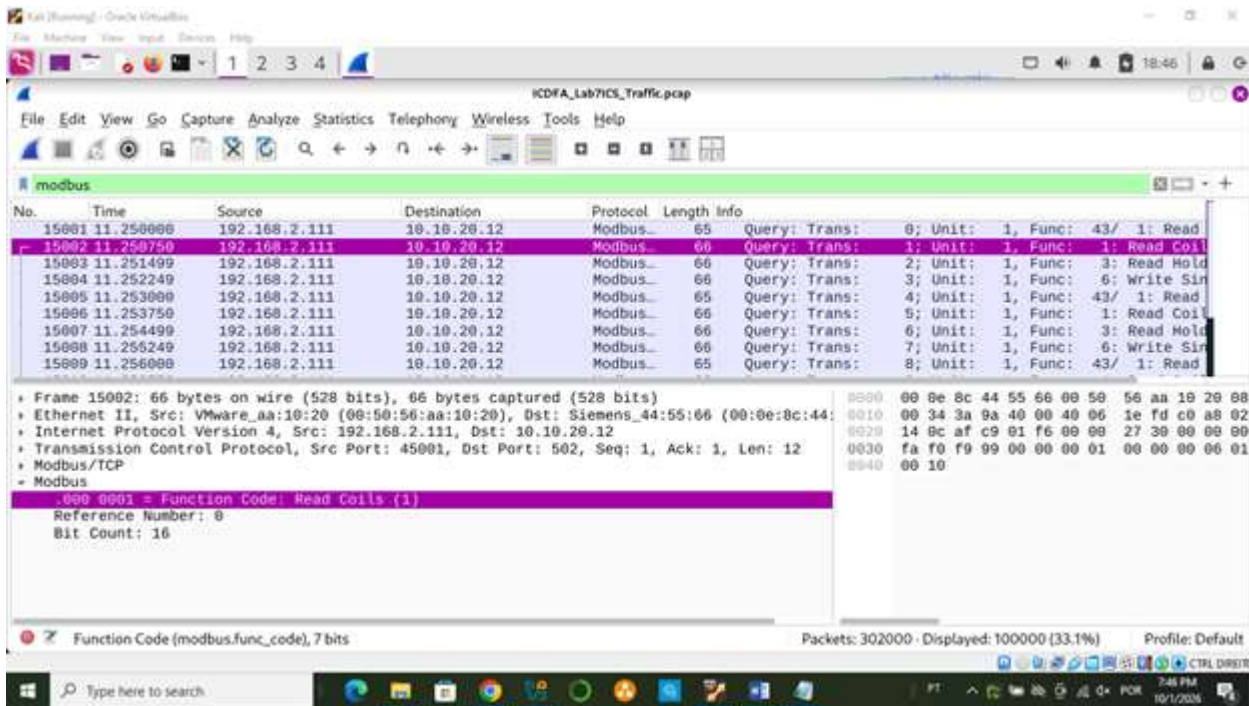
Step 5. Apply ftp and inspect the FTP session metadata. Record any product/vendor or software-version information visible in the capture.



Step 6. Apply modbus and identify at least two Modbus function codes. If the second teaching capture is supplied, open it and compare the volume/types of Modbus traffic.

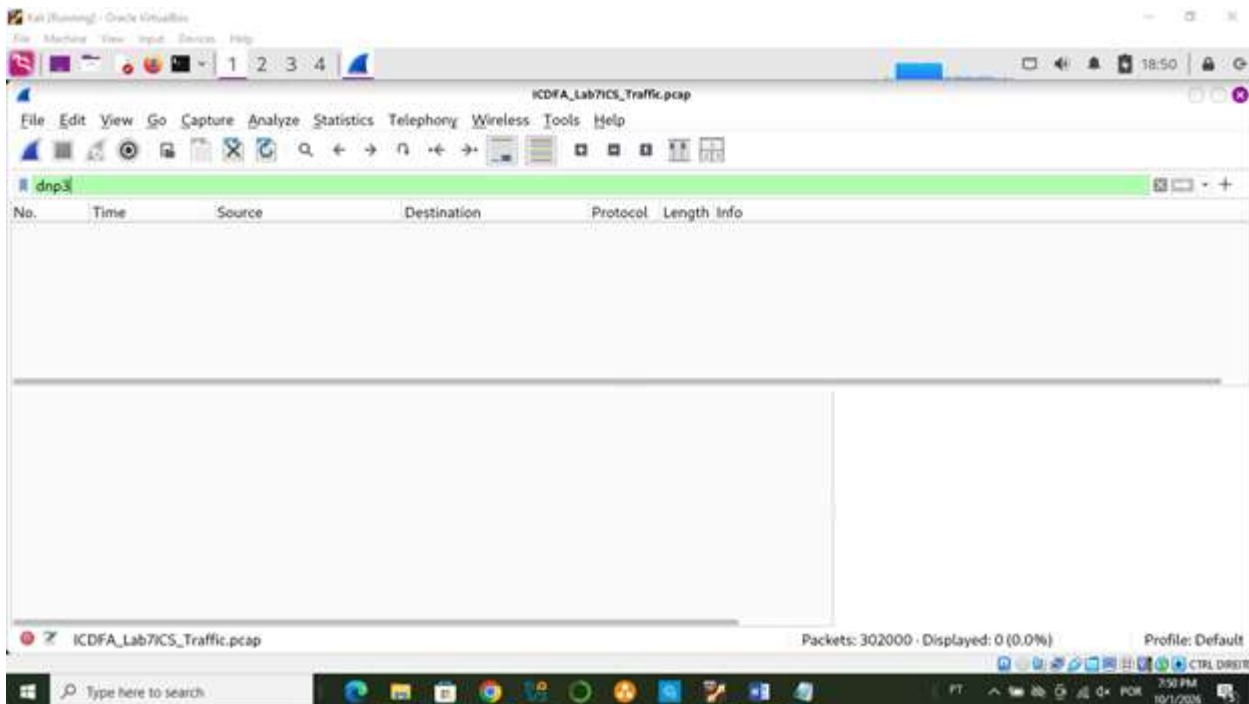


And

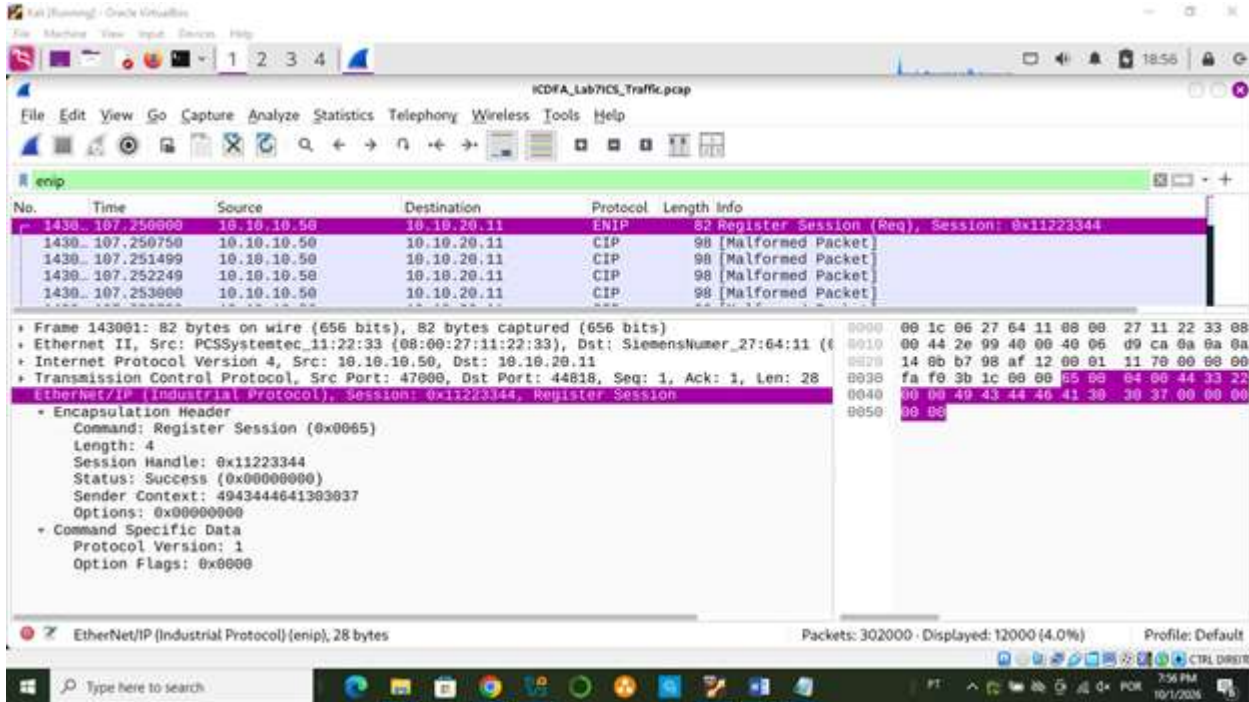


Step 7. Apply dnp3 and identify a DNP3 conversation. Record source/destination systems and explain why DNP3 is common in process automation/SCADA environments.

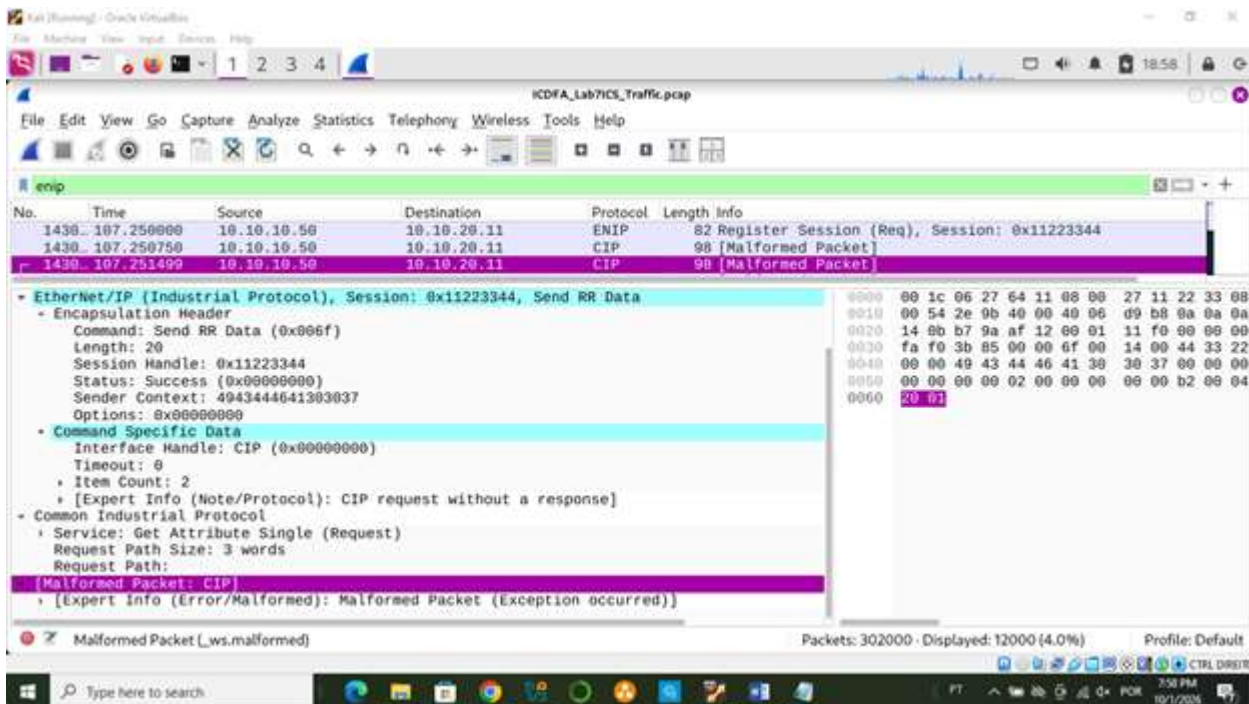
Remarks: There are no packets regarding DNP3 conversations



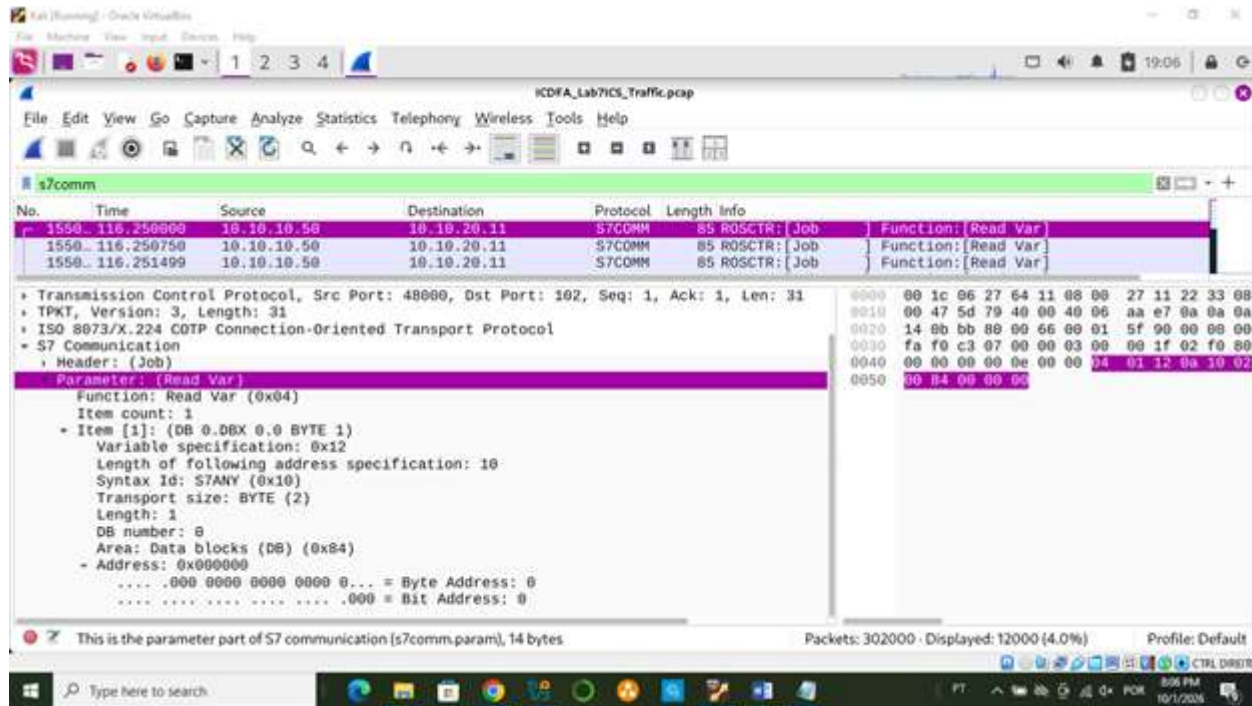
Step 8. Apply enip and inspect available EtherNet/IP/CIP traffic. Record the key header/protocol details visible in the capture.



And



Step 9. Apply s7comm and inspect an S7 communication exchange. Record source/destination information and any Read Var or comparable operation visible.



Step 10. Complete the Protocol Evidence Matrix for HTTP, FTP, Modbus, DNP3, EtherNet/IP and S7.

A: The table below summarizes the key indicators, filters, and evidence found within Wireshark packet details for each protocol:

Protocol	Standard Port	OSI Layer	Display Filters	Risk Evidence
HTTP	TCP 80	Application	http.authbasic http.request.method,	Data transmission in clear text, and credentials encoded in weak Base64 (Authorization: Basic); weak password & session token hash algorithm (MD5)
FTP	TCP 21	Application	ftp	Data transmission in clear text; lack of authentication & encryption (exposure of device metadata)
Modbus	TCP 502	Application (Used in ICS)	modbus	Data transmission in clear text; lack of authentication & encryption (Read Device Identification)
DNP3	TCP/UDP 20000	Application & Data Link Layer (Used in ICS)	dnp3	Data transmission in clear text; lack of authentication & encryption
Ethernet/IP	TCP/UDP 44818	Application (Used in ICS)	enip	Data transmission in clear text; lack of authentication & encryption
S7Comm	TPC 102	Application (Used in ICS)	S7comm	Data transmission in clear text; lack of authentication & encryption; exposed device metadata

Step 11. Write a security assessment identifying at least five observations such as cleartext/encoded credentials, legacy protocols, exposed device metadata, insufficient confidentiality or excessive network visibility.

A:

Security Assessment

Developed By: Pedro Miambo

Based on the network indicators mapped in the matrix above, following security issues have been identified:

- ✓ *Clear text and Weakly Encoded Credentials (HTTP/FTP):* The PCAP exposes credentials over unencrypted channels. FTP passes credentials in raw text, while HTTP wraps them in reversible Base64 encoding. Any threat actor with network sniffing tool can easily compromise these accounts.
- ✓ *Insufficient Confidentiality and Protocol Vulnerability (Modbus TCP):* The Modbus traffic demonstrates absence of encryption. A threat actor can send commands to and compromise integrity of physical field devices. In a real scenario, this would allow unauthorized users to manipulate production lines without triggering a protocol-level rejection.
- ✓ *Exposed Device Metadata and Asset Information (EtherNet/IP & S7 Comm):* By analyzing CIP identity requests (EtherNet/IP) and SZL system state lists (S7 Comm), the network leaks sensitive infrastructure details. This includes vendor names, PLC controller models, firmware versions and much more. A threat actor can use this data to target specific known vulnerabilities without being noticed.
- ✓ *Poor Network Segmentation:* Common office IT protocols (HTTP, FTP) mixed in the same packet capture file alongside highly sensitive industrial OT protocols (Modbus, DNP3, S7) points to a lack of functional network segmentation. The absence of a firewall boundary or DMZ means a breach in the corporate network can easily compromise the physical manufacturing or utility control zone.

Conclusion

The security analysis reveals critical risks due to a lack of network separation and the use of outdated, insecure systems. Because standard IT traffic and industrial protocols mix without encryption, a threat actor who enters the corporate network can easily move into the production zone. This allows gathering of sensitive data, steal of credentials, and directly control machinery or field devices, leading to production downtime, damage of equipment, and severe safety hazards for workers and environment.

To address these vulnerabilities, implementing a Defense-in-Depth strategy requires:

- ✓ Separating the network to isolate the production environment from the corporate network;
- ✓ Removing unencrypted protocols to stop data from being intercepted;
- ✓ Deploy secure, encrypted versions of industrial communication protocols.

Step 12. Save your Wireshark workspace/screenshots and submit the report. Do not use any discovered credentials, addresses or device information outside the provided training material.

Required Evidence & Submission

- At least six protocol-analysis screenshots, one for each required protocol family.
- A Protocol Evidence Matrix with filters, endpoints, observed operations and security significance.
- A one-page security assessment prioritizing the observed risks.
- Redacted evidence of any credential material; plaintext credentials must not appear in the submitted report.

Recommended filename: <RegistrationNo>_LAB07_Exploring_ICS_Packet_Captures.pdf

Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Response: Base64 is an encoding scheme that translates binary data into a limited set of ASCII characters, rather than an encryption algorithm.

2. What information can device banners or protocol metadata reveal to an analyst?

Response: Device banners or protocol metadata can reveal information such as hardware, software configurations, network architecture, and user behaviors without the need to examine the content of a communication.

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

Response: Protocols like Modbus, DNP3, EtherNet/IP, and S7 keep factories and critical infrastructure running in real time. They work very differently from the usual IT systems we use daily, like HTTP or FTP. While web protocols handle data downloads and browsing, these industrial systems are built strictly for fast machinery control, specific hardware data, and zero tolerance for downtime.

4. Why is passive packet analysis valuable for understanding an ICS environment?

Response: Passive packet analysis is valuable because it allows analysts to perform network monitoring without disrupting Operational Technology services. Since most of ICS devices have been designed to run on minimal computational resources, injecting high-volume traffic or active vulnerability scans can easily cause firmware crashes, buffer overflows, or unplanned reboots.

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Response: Since legacy industrial protocols do not provide features such as encryption and authentication, the following compensating controls can assist in reducing the risk: administrative & physical protections; network segmentation & isolation; access control & identity management and monitoring & defenses.

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.
Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.		